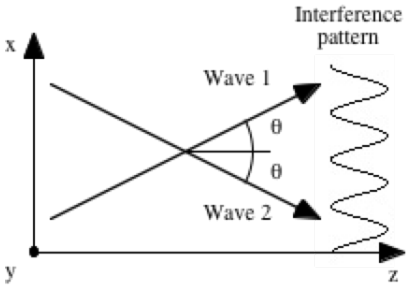
Optoelectronics 2022 – Solutions

1a) Assuming y-polarized waves propagating in free space in the $x-z$ plane at angles $\pm\theta$ to the z-axis, the electric field of the waves below can be written as $\tilde{E} = E_y\tilde{j}$ where:

$$E_y = E_1\exp\{-jk_0[z\cos(\theta) + x\sin(\theta)]\} + E_2\exp\{-jk_0[z\cos(\theta) - x\sin(\theta)]\}$$

[2]

Here $k_0 = 2\pi/\lambda$ is the propagation constant of free space.



If the wave amplitudes are equal (say, to E_0) we may write the above in modal form as:

$$E_y = E_0\{\exp\{-jk_0x\sin(\theta)\} + \exp\{+jk_0x\sin(\theta)\}\}\exp\{-jk_0z\cos(\theta)\}$$

Or:

$$E_y = 2E_0\cos\{k_0x\sin(\theta)\}\exp\{-jk_0z\cos(\theta)\}$$

[2]

b) This modal solution is an inhomogeneous wave. The transverse field variation is $E_y = 2E_0\cos\{k_0x\sin(\theta)\}$ and the peak amplitude is $2E_0$.

[2]

The field is travelling in the z direction and its effective propagation constant is $k_0\cos(\theta)$

[2]

c) The accompanying magnetic field can be found from the electric field using the Maxwell curl equation. For fields varying as $\exp(j\omega t)$, this can be written as $\nabla\times\tilde{E} = -j\omega\mu_0\tilde{H}$ so:

$$\tilde{H} = \left(\frac{j}{\omega\mu_0}\right)\nabla\times\tilde{E}$$

In this case, the electric field only has a y-component, so that:

$$\nabla\times\tilde{E} = \begin{vmatrix}\tilde{\imath} & \tilde{\jmath} & \tilde{k}\\ \dfrac{\partial}{\partial x} & \dfrac{\partial}{\partial y} & \dfrac{\partial}{\partial z}\\ 0 & E_y & 0\end{vmatrix} = -\frac{\partial E_y}{\partial z}\tilde{\imath} + \frac{\partial E_y}{\partial x}\tilde{k}$$

Hence, the magnetic field only has components:

$$H_x = \left(\frac{-j}{j\omega\mu_0}\right)\frac{\partial E_y}{\partial z} \text{ and } H_z = \left(\frac{+j}{\omega\mu_0}\right)\frac{\partial E_y}{\partial x}$$

[3]

Now the irradiance is:

$$\tilde{S} = \frac{1}{2}\mathrm{Re}\left(\tilde{E}\times\tilde{H}^*\right)$$

In this case:

$$\tilde{E}\times\tilde{H}^* = \begin{vmatrix}\tilde{\imath} & \tilde{\jmath} & \tilde{k}\\ 0 & E_y & 0\\ H_x^* & 0 & H_z^*\end{vmatrix} = E_yH_z^*\tilde{\imath} - E_yH_x^*\tilde{k}$$

On a screen perpendicular to the z-axis we will see the z-component of the irradiance:

Answers

$$S_z = \frac{1}{2} Re\left(-E_y H_x^*\right) = \frac{1}{2} Re\left(\frac{j}{\omega\mu_0} E_y \frac{\partial E_y^*}{\partial z}\right)$$

[3]

Doing the differentiation, we get:

$$\frac{\partial E_y}{\partial z} = -jk_0 \cos(\theta)\, E_y$$

So that:

$$\frac{\partial E_y^*}{\partial z} = jk_0 \cos(\theta) E_y^*$$

Hence:

$S_z = A \cos^2\{k_0 x \sin(\theta)\}$ with A= $2E_0^2 \frac{k_0 \cos(\theta)}{\omega\mu_0}$.

This is a periodic variation in intensity or an interference pattern.

[2]

d) The interference pattern repeats whenever $k_0 x \sin(\theta) = \nu\pi$, where ν is an integer. The period is therefore $\Lambda = \pi/\{k_0 \sin(\theta)\} = \lambda/\{2 \sin(\theta)\}$

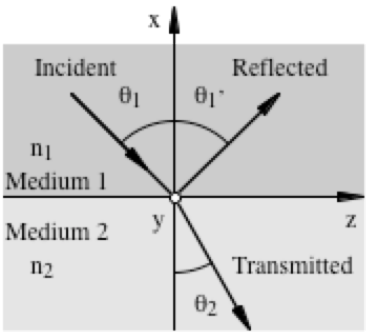
[2]

Assuming $\lambda = 0.5\mu m$ and $\theta = 30^o$, $\Lambda = 0.5/(2\ x\ 0.5) = 0.5\mu m$.

[2]

Answers

2. a) Assuming the wave is incident from the high index side, the ray diagram is:



[4]

b) The relation that must be satisfied by the angles of travel of the incident and transmitted wave is Snell's law, $n_1 \sin(\theta_1) = n_2 \sin(\theta_2)$.

[1]

Using Snell's Law, θ_2 can be found as $\theta_2 = \sin^{-1}\left\{\frac{n_1}{n_2}\sin(\theta_1)\right\}$

If $n_1 > n_2$, there may be an angle θ_{1C} at which $\frac{n_1}{n_2}\sin(\theta_{1C}) = 1$. Beyond this, there is no real solution for θ_2. The transmitted wave can no longer be a plane wave, and all the power is reflected. This is known as total internal reflection and the critical angle is $\theta_{1C} = \sin^{-1}(\frac{n_2}{n_1})$.

[3]

If $n_1 = 1.5$ and $n_2 = 1$, $\theta_{1C} = \sin^{-1}(\frac{1}{1.5}) = 41.81^o$

[1]

c) For y-polarization, the field in Medium 2 can be written before TIR as a plane wave:
$E_{y2} = E_2 \exp\{-jn_2k_0[z\sin(\theta_2) - x\cos(\theta_2)]\}$

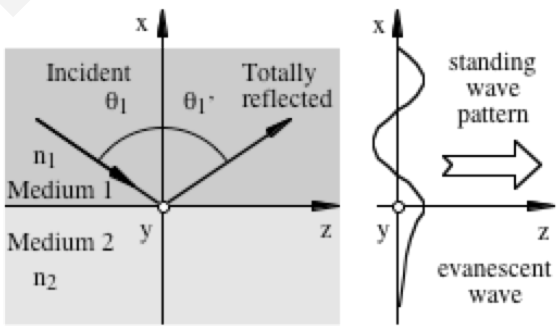
[2]

After TIR, $\sin(\theta_2) = \frac{n_1}{n_2}\sin(\theta_1)$ is real, but greater than unity.

Consequently $\cos(\theta_2) = \sqrt{1 - \sin^2(\theta_2)}$ must be purely imaginary, so $\cos(\theta_2) = \pm j\alpha$, say. Keeping the sign that leads to exponential decay below the interface, the field becomes: $E_{y2} = E_2 \exp\{-jn_2k_0\, z\sin(\theta_2)\}\exp(\alpha x)$. This is an evanescent wave.

[2]

The field in Medium 1 is the sum of two intersecting waves, i.e. a standing wave pattern, which joins smoothly on to the evanescent field as shown below.



[2]

Answers

d) The loss mechanisms operating in the three spectral regions are:
A – Rayleigh scattering
B – Hydroxyl (OH^-) ion absorption
C – Mid IR molecular vibrational absorption

[3]

The optimum operating wavelength is the point at which the Rayleigh scattering and molecular absorption envelopes cross, at around $\lambda = 1.55\ \mu m$.

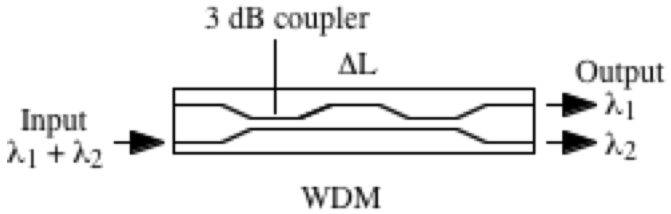
[1]

At this point the loss is around 0.2 dB/km.

[1]

Answers

3. a) The filter can be analysed by summing the mode amplitudes obtained via all possible paths between the input and the two outputs, ignoring the common phase delay.



We assume that the bar-state and cross-state outputs of a 3 dB coupler are $1/\sqrt{2}$ and $-j/\sqrt{2}$, the propagation is constant is β and the imbalance is ΔL. Summing the amplitudes of the two paths between the lower input and lower output gives:

$$A_1 = \frac{1}{\sqrt{2}} x \frac{1}{\sqrt{2}} + \frac{-j}{\sqrt{2}} \exp(-j\beta\Delta L) x \frac{-j}{\sqrt{2}}$$

Summing the amplitudes of the two paths between the lower input and upper output gives:

$$A_2 = \frac{1}{\sqrt{2}} x \frac{-j}{\sqrt{2}} + \frac{-j}{\sqrt{2}} \exp(-j\beta\Delta L) x \frac{1}{\sqrt{2}}$$

[4]

Multiplying out and re-arranging, we get:

$$A_1 = \frac{1}{2}\{1 - \exp(-j\beta\Delta L)\}$$
$$A_2 = \frac{-j}{2}\{1 + \exp(-j\beta\Delta L)\}$$

We can write this as:

$$A_1 = \frac{1}{2}\{\exp(+j\beta\Delta L/2) - \exp(-j\beta\Delta L/2)\}\exp(-j\beta\Delta L/2)$$
$$A_2 = \frac{-j}{2}\{\exp(+j\beta\Delta L/2) + \exp(-j\beta\Delta L/2)\}\exp(-j\beta\Delta L/2)$$

So that:

$$A_1 = +j\sin(\beta\Delta L/2)\exp(-j\beta\Delta L/2)$$
$$A_2 = -j\cos(\beta\Delta L/2)\exp(-j\beta\Delta L/2)$$

The two output powers are then:

$$P_1 = |A_1|^2 = \sin^2(\beta\Delta L/2)$$
$$P_2 = |A_2|^2 = \cos^2(\beta\Delta L/2)$$

[4]

b) The spectral variation can be found as follows. The propagation constant is $\beta = \frac{2\pi n_{eff}}{\lambda}$. If we write $\lambda = \lambda_0 + \delta\lambda$, where $\delta\lambda$ is small, then:

$$\beta = \left(\frac{2\pi n_{eff}}{\lambda_0}\right)\left(\frac{1}{1 + \delta\lambda/\lambda_0}\right) \approx \left(\frac{2\pi n_{eff}}{\lambda_0}\right)(1 - \delta\lambda/\lambda_0)$$

Hence:

$$\beta\Delta L \approx \beta_0\Delta L(1 - \delta\lambda/\lambda_0)$$

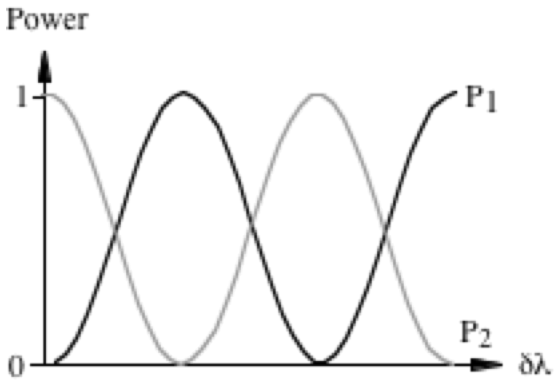
[2]

The power outputs must then vary sinusoidally with $\delta\lambda$ as shown below.

The problem with MZI-based filters is that it is difficult to choose ΔL to drop two desired values of λ from the two ports, and even harder to design a cascaded system capable of separating the multiple wavelengths in a WDM communication system.

Answers

[2]

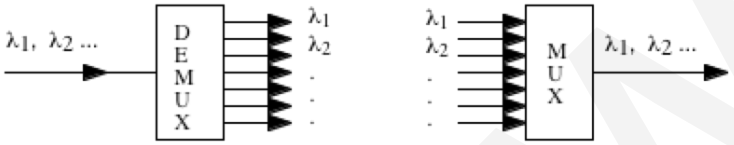


[2]

c) The arrayed waveguide grating multiplexer forms the basic building block of WDM systems, since it provides a bank of near-identical filters with regularly spaced centre frequencies using a single compact device.

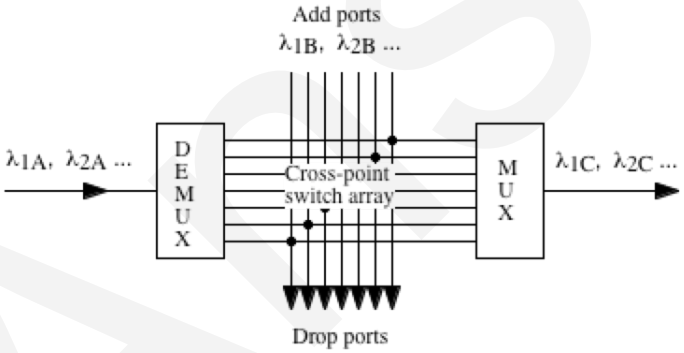
[2]

The AWG MUX is reversible, so it can act as a DEMUX. Block diagrams are as below:



[2]

Combination of a pair of AWG MUX components and a cross-point optical switch array provides an ADD-DROP MUX as below:



[2]

Answers

4. a) The coupled mode equations for a synchronous directional coupler can be written as:

$$\frac{dA_1}{dx} + j\kappa A_2 = 0$$
$$\frac{dA_2}{dx} + j\kappa A_1 = 0$$

Here A_1 and A_2 are the amplitudes of modes in Guides 1 and 2, κ is the coupling coefficient and x is the distance along the device.

[2]

The coupling coefficient depends on the overlap of the transverse electric field in Guide i with the refractive index perturbation introduced by Guide j. Since the transverse field decays exponentially outside each guide, κ typically decays exponentially with guide separation.

[2]

b) Differentiating the upper equation, we can obtain:

$$\frac{d^2A_1}{dx^2} + j\kappa\frac{dA_2}{dx} = 0$$

Substituting using the lower equation, we get:

$$\frac{d^2A_1}{dx^2} + \kappa^2 A_1 = 0$$

Solutions are of the firm:

$$A_1 = A\cos(\kappa x) + B\sin(\kappa x)$$

So that:

$$\frac{dA_1}{dx} = -j\kappa A\sin(\kappa x) + j\kappa B\cos(\kappa x)$$

[2]

Boundary conditions for unity input to Guide 1 are $A_1|_0 = 1, A_2|_0 = 0$. Using the upper equation, these can be rewritten as: $A_1|_0 = 1, \frac{dA_1}{dx}|_0 = 0$ and imply that $A = 1, B = 0$.

[2]

Hence:

$$A_1 = \cos(\kappa x)\,;\; A_2 = \left(\frac{-1}{j\kappa}\right)\frac{dA_1}{dx} = -j\sin(\kappa x)$$

Hence the powers vary as $P_1 = |A_1|^2 = \cos^2(\kappa x)\,;\; P_2 = |A_2|^2 = \sin^2(\kappa x)$

[2]

For a 3 dB power splitter, we require $P_1 = P_2 = 1/2$, so that $\kappa L = \frac{\pi}{4}$ in a device of length L.

[2]

c) If the coupling coefficient varies exponentially with guide separation g, we may write:

$$\kappa = A\exp(-Bg)$$

Here A and B are two unknown constants. Their values may be found as follows. If the coupling coefficient obtained for a guide separation g_1 is κ_1 and the corresponding value for a separation g_2 is κ_2, we may then write:

$$\kappa_1 = A\exp(-Bg_1)$$
$$\kappa_2 = A\exp(-Bg_2)$$

Hence:

$$\frac{\kappa_2}{\kappa_1} = \exp\{-B(g_2 - g_1)\}$$

If the corresponding lengths required for 3 dB couplers are L_1 and L_2 then:

Answers

$$\frac{L_1}{L_2} = \exp\{-B(g_2 - g_1)\}$$

Re-arranging:

$$B = \frac{-1}{(g_2 - g_1)} \log\left(\frac{L_1}{L_2}\right)$$

[2]

If $L_1 = 2e^{-3}$ for $g_1 = 3e^{-6}$ and $L_2 = 3e^{-3}$ for $g_2 = 4e^{-6}$ then:

$$B = \frac{-1e^{6}}{(4-3)} \log\left(\frac{2}{3}\right) = 4.0547e^{5}$$

[2]

Substituting, we can find A as:

$$A = \kappa_1 \exp(Bg_1) = \left(\frac{\pi}{4L_1}\right) \exp(Bg_1)$$

Hence:

$$A = \left(\frac{\pi}{4x2e^{-3}}\right) \exp(4.0547e^{5}x3e^{-6}) = 1.3254e^{3}$$

Alternatively, we could use:

$$A = \left(\frac{\pi}{4x3e^{-3}}\right) \exp(4.0547e^{5}x4e^{-6}) = 1.3254e^{3}$$

However, the result must clearly be the same.

[2]

Finally, for a gap g_3 we obtain:

$$\kappa_3 = A \exp(-Bg_3)$$

If $g_3 = 5\mu m$ we then get:

$$\kappa_3 = 1.3254e^{3} \exp(-4.0547e^{5}x5e^{-6}) = 174.5m^{-1}$$

The corresponding length is then:

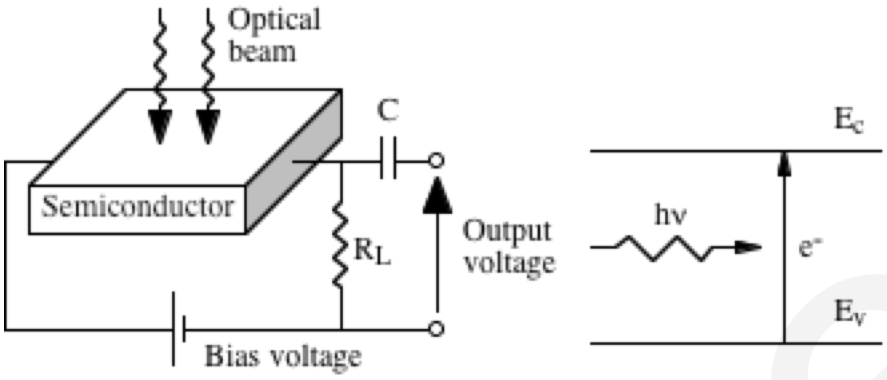
$$L_3 = \frac{\pi}{4\kappa_3} = \frac{\pi}{4x174.5} = 0.0045, \text{or } 4.5\ mm$$

[2]

Answers

5. a) A photoconductive detector operates by the promotion of electrons from the valence band to the conduction band in a semiconductor. Assume that light beam illuminates a slab of semiconductor. Photons may only be absorbed if their energy $E = hf = hc/\lambda$ satisfies $E > E_g$ where $E_g = E_c - E_v$. Each absorbed photon creates an electron-hole pair, increasing the conductivity. The change in conductivity is proportional to the beam power and is detected by arranging the photoconductor in a potential divider with a load resistor R_L.

[2]

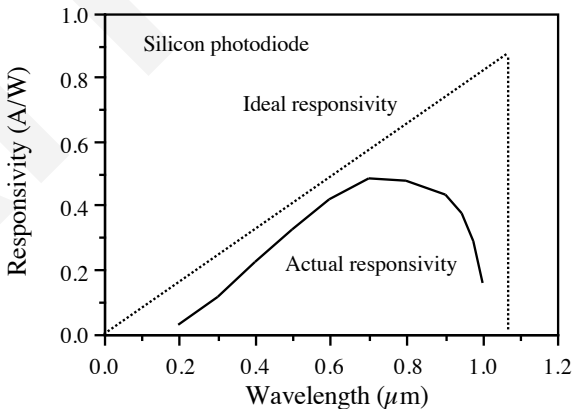


[2]

b) To derive the responsivity, assume a beam of power P falls on the detector
If each photon has energy $hf = hc/\lambda$, their rate of arrival is $P/hf = P\lambda/hc$ (sec^{-1})
Assume each photon generates one electron transiting the detector
The rate of generation of electrons is then also $P\lambda/hc$
Assume each electron carries a charge e; the ideal photocurrent is then $I_P = Pe\lambda/hc$ (A)
Not all photons generate an electron; assume the fraction that do is η (quantum efficiency)
The actual photocurrent is then $I_P = P\eta e\lambda/hc$ (A)
The responsivity (current per unit optical power) is $R = \frac{I_P}{P} = \eta e\lambda/hc$ (A/W)

[4]

The ideal spectral variation in responsivity with $\eta = 1$ is then the dotted straight line below, which continues until a maximum wavelength set by the energy gap, namely $\lambda_m = hc/eE_g$.



[2]

c) if $\eta = 1$, at $\lambda = 1\ \mu m$ we get $R = 1.6e^{-19}x1e^{-6}/(6.62e^{-34}x3e^{8}) = 0.8056\ A/W$

[2]

Answers

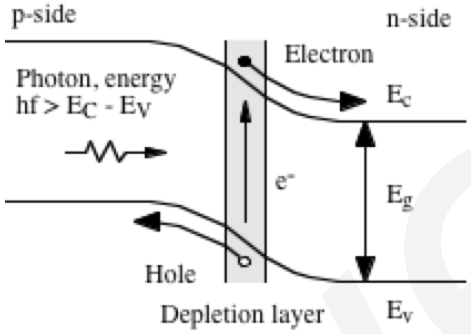
At $\lambda = 2\ \mu m$ wavelength, $R = 0$, since $\lambda_m = \frac{6.62e^{-34}x3e^{8}}{(1.6e^{-19}x1.14)} m = 1.088\ \mu m.$

[2]

d) One disadvantage is that the conductivity will not be zero when beam is off (due to thermally generated carriers) so there will always be a DC voltage across R_L. The DC voltage may be blocked by a capacitor C. However, thermal carriers generate noise, and this cannot be blocked. Another disadvantage is that long lifetime is needed for high sensitivity, but carriers then persist after beam is switched off, so it is difficult to combine high sensitivity with high speed.

[2]

A solution is provided by making absorption occur in the depletion region of p-n-junction diode, since the carrier pairs can easily be separated by the large built-in field there and can reach the contacts without recombining. Photodiodes are therefore both fast and efficient.



[2]

The responsivity of a practical photodiode is shown as the full line in the earlier diagram. The quantum efficiency is reduced by absorption of high energy photons before they reach the depletion layer (at short wavelengths) and absorption of low energy after they have passed through it (at long wavelengths).

[2]

Answers

6. The rate equations for a light-emitting diode can be written in the form:

$$\frac{dn}{dt} = \frac{I}{ev} - \frac{n}{\tau_e}$$
$$\frac{d\phi}{dt} = \frac{n}{\tau_{rr}} - \frac{\phi}{\tau_p}$$

a) Terms in the RHS above are as follows:

$\frac{I}{ev}$ – Increase in electron density due to current injection

$\frac{n}{\tau_e}$ – Decrease in electron density due to all types of recombination

$\frac{n}{\tau_{rr}}$ – Increase in photon density due to radiative recombination

$\frac{\phi}{\tau_p}$ – Decrease in photon density due to departure from the active volume

[2]

The upper equation contains τ_e while the lower equation contains τ_{rr} because electrons can be lost by non-radiative and radiative radiation while only the latter generates photons.

[1]

The term ϕ/τ_p (the rate of escape of photons per unit volume) is of interest because it can be used to determine the output power, while ϕ (the photon density in the material) is an essentially unmeasurable quantity.

[1]

b) In the steady state the LHS is zero and the equations reduce to:

$$\frac{n}{\tau_e} = \frac{I}{ev}$$
$$\frac{\phi}{\tau_p} = \frac{n}{\tau_{rr}} = \left(\frac{\tau_e}{\tau_{rr}}\right)\frac{I}{ev}$$

Here the term $\frac{\tau_e}{\tau_{rr}}$ is the internal efficiency η_i.

[2]

The total photon flux out is found by multiplying by the active volume v, as $\frac{v\phi}{\tau_p} = \eta_i \frac{I}{e}$.

Assuming each photon carries an energy $hf = \frac{hc}{\lambda}$, the power output is $P = \eta_i \frac{I}{e}\left(\frac{hc}{\lambda}\right)$

[2]

The electron recombination lifetime is related to the radiative and non-radiative recombination lifetimes as $\frac{1}{\tau_e} = \frac{1}{\tau_{rr}} + \frac{1}{\tau_{nr}}$. If $\tau_{rr} = \tau_{nr} = 1ns$, $\tau_e = 0.5ns$ and $\eta_i = 0.5$.

[1]

The power output in $\frac{Watts}{Amp}$ is then $\frac{P}{I} = \eta_i\left(\frac{hc}{e\lambda}\right)$ or $\frac{P}{I} = 0.5\ x\ \frac{6.62e^{-34}x3e^{8}}{1.6e^{-19}x0.85e^{-6}} = 0.73mW/mA$

[1]

c) External efficiency is limited by total internal reflection at the semiconductor/air interface.

[1]

η_e can be estimated as follows. The critical angle at the interface between a medium of refractive index n and air is $\theta_C = sin^{-1}\left(\frac{1}{n}\right) \approx \frac{1}{n}$

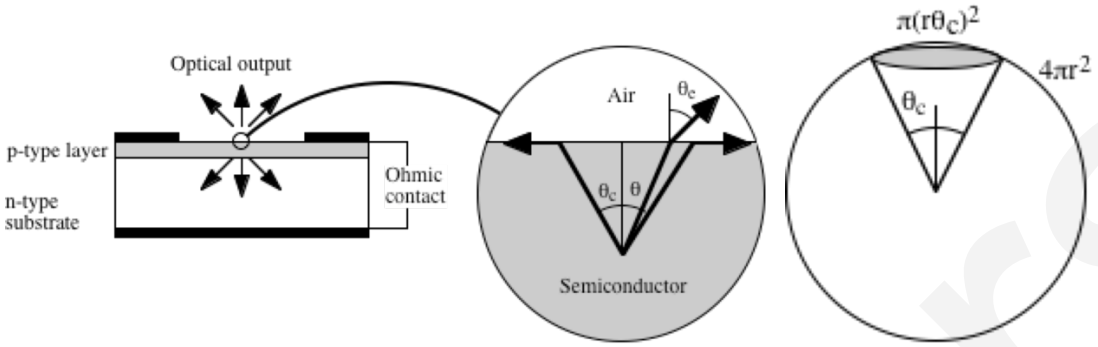
Assuming that radiation is generated isotropically, the fraction of power within the critical angle can be estimated from the diagram below as $F \approx \frac{\pi(\theta_C r)^2}{4\pi r^2} \approx \frac{1}{4n^2}$

Answers

The fraction of power transmitted across the interface is $T^2 = 1 - R^2$, where R is the reflection coefficient. Since $R = \frac{1-n}{1+n}$ for normal incidence, $T^2 \approx \frac{4n}{(1+n)^2}$

Hence, the external efficiency is $\eta_e \approx T^2 F = \frac{1}{n(1+n)^2}$

[3]



[1]

If $n = 3.5$, $\eta_e \approx \frac{1}{3.5(4.5)^2} = 0.014$, or 1.4%

[1]

d) LEDs for lighting require

1. Convoluted rather than flat surface to increase external efficiency
2. Multiple emitters to achieve sufficient brightness
3. Multiple emitters at different wavelengths to achieve white light.
4. Additional circuit to convert AC supply to low-voltage DC

[2]

Due to increased generation of photons rather than phonons, efficiency higher is much higher than filament light bulbs or fluorescent lights. Due to the lower operating temperature, life expectancy is much longer. Energy and servicing costs are therefore much lower. However, the initial capital cost is higher.

[2]

Answers